

Process	Unit Cost	Unit
Die Casting	\$ 4.00	kg
Cold Forging, per Step	\$ 2.50	kg
Hot Forging, per Step	\$ 3.00	kg
Investment Casting	\$ 8.00	kg
Liquid fill molding		kg
Lost Foam Casting	\$ 3.50	kg
Plastic injection molding	\$ 2.75	kg
Powder Metal Forming	\$ 3.00	kg
Profile Extrusion	\$ 3.00	kg
Rapid Prototype - Composite	\$ 175.00	kg
Rapid Prototype - Metal	\$ 330.00	kg
Rapid Prototype - Plastic	\$ 32.00	kg
Sand Casting	\$ 3.00	kg
Vacuum Form	\$ 10.00	m^2
Anodizing, Cosmetic	\$ 0.00	cm^2
Anodizing, Hardened	\$ 0.01	cm^2
Cosmetic Chrome Plating	\$ 40.00	m^2
Exhaust System Ceramic Coating	\$ 25.00	m
Galvanize Coating	\$ -	
Hard Chrome Plating	\$ 50.00	m^2
Hardened Electroless Nickel Plating	\$ 20.00	
Spray Insulation Application, UL94 Rated	\$ 20.00	m^2
Hand/Machine Finish - Surface preparation for painting	\$ 10.00	m^2
Nitriding/Case Hardening	\$ 10.00	m^2
Paint Apply, Schultz Type Application	\$ 2.00	m^2
Paint Apply, Pressurized Gun or Hand Application	\$ 5.00	m^2
Cure, Autoclave	\$ 50.00	m^2
Cure, Oven	\$ 20.00	m^2
Cure, Room Temperature, Vacuum	\$ 10.00	m^2
Cure, Room Temperature, Atmospheric	\$ -	m^2
Lamination, Filament Winding	\$ 25.00	kg
Lamination, Forged Composite	\$ 25.00	kg
Lamination, Manual	\$ 35.00	m^2
Potting	\$ 0.50	cm
Resin application, Infusion Molding	\$ 2.50	m^2
Resin application, Manual	\$ 5.00	m^2
Circuit Card Assembly Labor - Hand Soldering	\$ 0.05	unit
Circuit Card Assembly Labor - Pick and Place	\$ 0.01	unit
Circuit Card Assembly Labor - Reflow Oven	\$ 1.00	unit
Conformal Coat Apply	\$ 5.00	unit
Solder Paste Apply	\$ 1.00	unit
Cut wire	\$ 0.08	unit
Strip Wire	\$ 0.08	unit

Tin Wire	\$	0.13	unit
Crimp Wire	\$	0.17	unit
Strip Multi-Conductor	\$	0.13	wire
Connector Assembly, Crimp	\$	0.36	contact
Connector Assembly, Solder	\$	0.24	contact
Shrink Tube	\$	0.15	cm
Lay Wire	\$	0.02	m
Taping Wire Bundle	\$	0.04	cm
Lace	\$	0.15	unit
Insert Bundle Into Tube or Sleeve	\$	0.02	m
Install Adhesive Cable Clamp	\$	0.19	unit
Wire Dressing (Install and route)	\$	1.00	m
Install Tie Wrap (Zip Tie, Cable Clamp)	\$	0.09	unit
Connector Install	\$	0.11	unit
Connector Install, Individual Quick Connect	\$	0.10	unit
Connector Install, With Fasteners	\$	0.50	unit
Attach Wire, Ring	\$	0.48	unit
Attach Wire, Solder	\$	0.43	wire
Attach Wire, to Fastener or Terminal Block	\$	0.50	wire
Hand - Start Only	\$	0.12	unit
Hand, Loose <= 25.4 mm	\$	0.50	unit
Hand, Loose <= 6.35 mm	\$	0.25	unit
Hand, Loose > 25.4 mm	\$	0.75	unit
Hand, Tight <= 6.35 mm	\$	0.50	unit
Power Tool <= 25.4 mm	\$	0.25	unit
Power Tool <= 6.35 mm	\$	0.25	unit
Power Tool > 25.4 mm	\$	0.50	unit
Ratchet <= 25.4 mm	\$	0.75	unit
Ratchet <= 6.35 mm	\$	0.50	unit
Ratchet > 25.4 mm	\$	1.50	unit
Reaction Tool <= 25.4 mm	\$	0.25	unit
Reaction Tool <= 6.35 mm	\$	0.25	unit
Reaction Tool > 25.4 mm	\$	0.50	unit
Screwdriver < 1 Turn	\$	0.12	unit
Screwdriver > 1 Turn	\$	0.50	unit
Tensioning Tool <=6.35 mm	\$	0.25	unit
Tensioning Tool <=25.4 mm	\$	0.25	unit
Tensioning Tool >25.4 mm	\$	0.50	unit
Wrench <= 25.4 mm	\$	1.50	unit
Wrench <= 6.35 mm	\$	1.00	unit
Wrench > 25.4 mm	\$	2.00	unit
Braze	\$	0.15	cm

Sewing	\$	0.08	cm
Weld	\$	0.15	cm
Aerosol Apply	\$	5.25	m^2
Brush Apply	\$	0.00	cm^2
Liquid Applicator Gun	\$	0.02	cm
Liquid Apply - Spot	\$	0.10	unit
Assemble, >20 kg, Interference	\$	5.63	unit
Assemble, >20 kg, Line-on-Line	\$	3.75	unit
Assemble, >20 kg, Loose	\$	1.88	unit
Assemble, 1 kg, Interference	\$	0.19	unit
Assemble, 1 kg, Line-on-Line	\$	0.13	unit
Assemble, 1 kg, Loose	\$	0.06	unit
Assemble, 10 kg, Interference	\$	1.88	unit
Assemble, 10 kg, Line-on-Line	\$	1.25	unit
Assemble, 10 kg, Loose	\$	0.63	unit
Assemble, 15 kg, Interference	\$	2.81	unit
Assemble, 15 kg, Line-on-Line	\$	1.88	unit
Assemble, 15 kg, Loose	\$	0.94	unit
Assemble, 20 kg, Interference	\$	3.75	unit
Assemble, 20 kg, Line-on-Line	\$	2.50	unit
Assemble, 20 kg, Loose	\$	1.25	unit
Assemble, 3 kg, Interference	\$	0.56	unit
Assemble, 3 kg, Line-on-Line	\$	0.38	unit
Assemble, 3 kg, Loose	\$	0.19	unit
Assemble, 5 kg, Interference	\$	0.94	unit
Assemble, 5 kg, Line-on-Line	\$	0.63	unit
Assemble, 5 kg, Loose	\$	0.31	unit
Tubing/Pipe Dressing (Install and route), Flexible Line	\$	1.00	m
Adjustment - Misc.	\$	5.00	unit
Brake Bleed - Per Bleeder Valve	\$	2.50	unit
Engine first start, includes fuel	\$	50.00	unit
Programming, Dataset Upload, End of Line	\$	5.00	Unit
Programming, Calibration	\$	10.00	Unit
Suspension Setup - Solid Axle (per corner)	\$	4.50	unit
Suspension Setup-Independent Susp. (per corner)	\$	8.75	unit
Cut (scissors, knife)	\$	0.06	cm
Heat Shrink, Fabric Panel	\$	5.00	m^2
Heat Treatment	\$	-	unit
Liquid Apply - Pour Expanding Foam	\$	2.00	m^3
Machining Setup, Change	\$	0.65	unit
Machining Setup, Install and remove	\$	1.30	unit
Riveting	\$	0.25	unit

Safety Wire, Install	\$	0.60	unit
Tape	\$	0.80	m
Vinyl Wrap Application	\$	10.00	m^2
Broach, External	\$	0.50	cm
Broach, Internal	\$	0.50	cm
Drilled hole < 50.8 mm dia.	\$	0.70	hole
Drilled holes < 25.4 mm dia.	\$	0.35	hole
EDM - Plunge	\$	0.30	cm^3
EDM - Wire, 2 Axis	\$	0.20	cm
EDM - Wire, Multiaxis	\$	0.00	cm^2
Gear Shaping (hobbing)	\$	0.50	cm
Grind, Cylindrical	\$	0.15	cm^2
Grind, Flat	\$	0.15	cm^2
Grind, Profile	\$	0.15	cm^2
Hand Finish - Material Removal	\$	0.20	cm^3
Hand Finish - Surface Preparation	\$	0.20	cm^2
Hot Wire Sculpting, 2 Axis	\$	0.10	cm
Hot Wire Sculpting, Multiaxis	\$	0.00	cm^2
Knurling	\$	0.10	cm
Laser Cut	\$	0.01	cm
Machining	\$	0.04	cm^3
Non-metallic cutting	\$	0.14	cm
Plasma Cutting	\$	0.10	cm
Reamed hole	\$	0.35	hole
Tapping holes	\$	0.35	hole
Threading, External (machining)	\$	0.10	cm
Threading, Internal (machining)	\$	0.10	cm
Waterjet Cut	\$	0.01	cm
Waterjet Cut, >50mm thick	\$	0.02	cm
Spinning, Metal	\$	0.02	cm^3
Threading, External (Rolling)	\$	0.10	unit
Sheet metal bends	\$	0.25	bend
Sheet metal punching	\$	0.03	cm^2
Sheet Metal Saw Cut	\$	0.20	cm
Sheet metal shearing	\$	0.25	cut
Sheet metal stamping	\$	0.03	cm^2
Tube bends	\$	0.75	bend
Tube cut	\$	0.15	cm
Tube end preparation for welding	\$	0.75	end
Tube Flare/Bead	\$	0.45	end
Tubing Lasercut Profiling/end preparation	\$	0.04	cm